

```

Started ossec-syscheckd...
Started ossec-monitord...
Completed.
hong@HIDS-OSSEC:~/ossec-hids-3.7.0$ sudo /var/ossec/bin/agent_control -lc

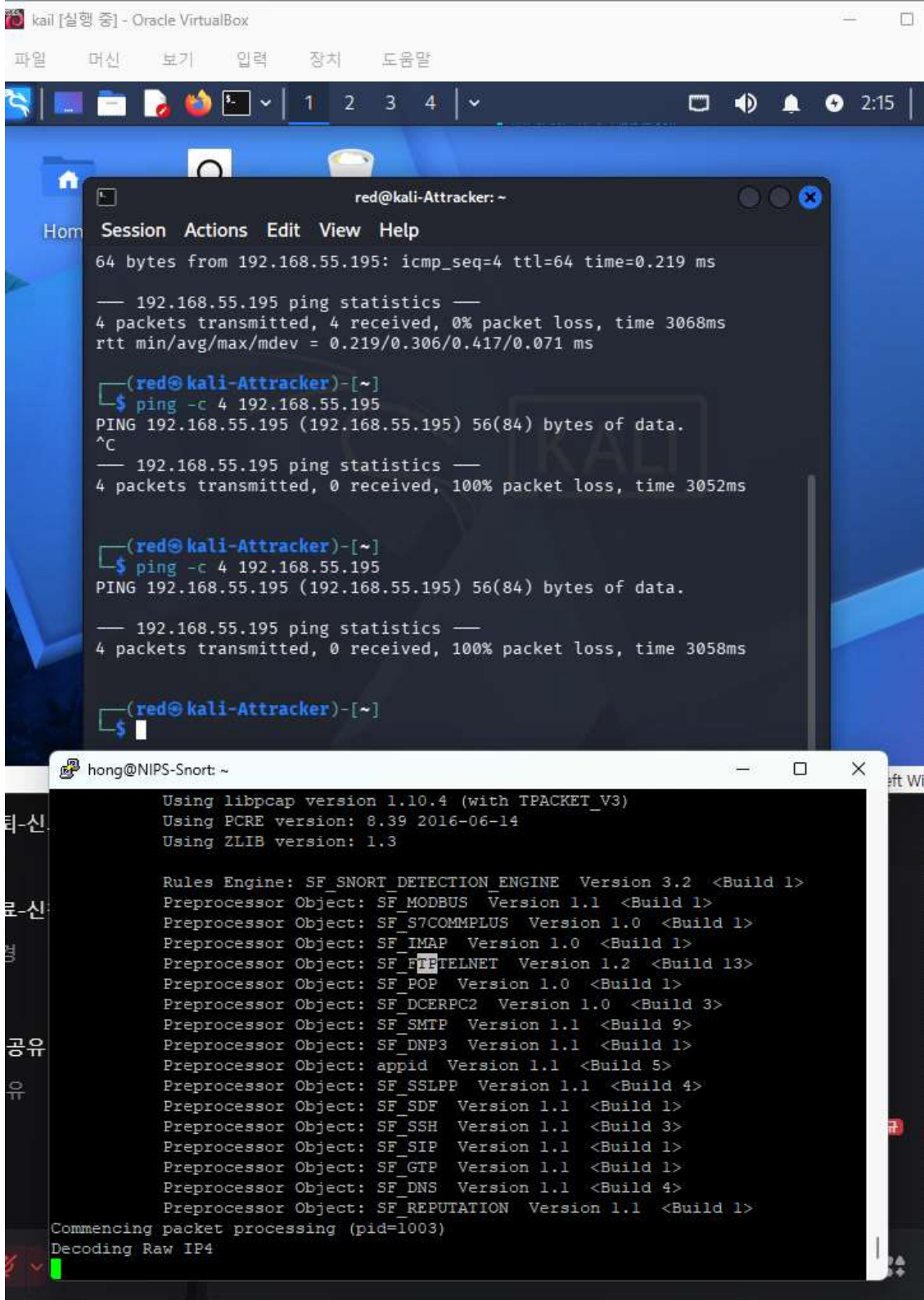
OSSEC HIDS agent_control. List of available agents:
  ID: 000, Name: HIDS-OSSEC (server), IP: 127.0.0.1, Active/Local
  ID: 001, Name: DVWA, IP: 192.168.55.240, Active

hong@HIDS-OSSEC:~/ossec-hids-3.7.0$ ^C
hong@HIDS-OSSEC:~/ossec-hids-3.7.0$ touch setuid_test.sh
hong@HIDS-OSSEC:~/ossec-hids-3.7.0$ 3.7.0$ sudo /var/ossec/bin/agent_control -lc
3.7.0$: command not found
hong@HIDS-OSSEC:~/ossec-hids-3.7.0$ ls -l setuid_test.sh
-rw-rw-r-- 1 hong hong 0 May 15 05:40 setuid_test.sh
hong@HIDS-OSSEC:~/ossec-hids-3.7.0$ chmod 4755 setuid_test.sh
hong@HIDS-OSSEC:~/ossec-hids-3.7.0$ ls -l setuid_test.sh
-rwsr-xr-x 1 hong hong 0 May 15 05:40 setuid_test.sh
hong@HIDS-OSSEC:~/ossec-hids-3.7.0$ ^C
hong@HIDS-OSSEC:~/ossec-hids-3.7.0$ mkdir sticky_test_dir
hong@HIDS-OSSEC:~/ossec-hids-3.7.0$ chmod 1777 sticky_test_dir
hong@HIDS-OSSEC:~/ossec-hids-3.7.0$ ls -ld sticky_test_dir
drwxrwxrwt 2 hong hong 4096 May 15 05:42 sticky_test_dir
hong@HIDS-OSSEC:~/ossec-hids-3.7.0$

```

사용한 명령어 /var/ossec/bin/agent_control -lc (에이전트 연결 확인) chmod 4755 setuid_test.sh (SetUID 설정) , chmod 1777 sticky_test_dir (Sticky Bit 설정)

설명:OSSEC HIDS 서버를 구축하고 DVWA 리눅스 에이전트를 연결하여 Active 상태를 확인했습니다. 이후 시스템 보안 실습으로 특정 스크립트에 SetUID를 부여하여 일반 사용자가 루트 권한으로 실행할 수 있게 설정하였고, 공유 디렉터리에 Sticky Bit를 설정하여 소유자만 파일을 삭제할 수 있도록 제한하였습니다.



```
kali [실행 중] - Oracle VirtualBox
파일  머신  보기  입력  장치  도움말
1  2  3  4
red@kali-Attracker: ~
Session Actions Edit View Help
PING 192.168.55.249 (192.168.55.249) 56(84) bytes of data.
64 bytes from 192.168.55.249: icmp_seq=1 ttl=64 time=0.258 ms
64 bytes from 192.168.55.249: icmp_seq=2 ttl=64 time=0.234 ms
64 bytes from 192.168.55.249: icmp_seq=3 ttl=64 time=0.318 ms
64 bytes from 192.168.55.249: icmp_seq=4 ttl=64 time=0.412 ms

— 192.168.55.249 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3057ms
rtt min/avg/max/mdev = 0.234/0.305/0.412/0.068 ms

(red@kali-Attracker)-[~]
$ ping -c 4 192.168.55.249
PING 192.168.55.249 (192.168.55.249) 56(84) bytes of data.
64 bytes from 192.168.55.249: icmp_seq=1 ttl=64 time=0.224 ms
64 bytes from 192.168.55.249: icmp_seq=2 ttl=64 time=0.223 ms
64 bytes from 192.168.55.249: icmp_seq=3 ttl=64 time=0.256 ms
64 bytes from 192.168.55.249: icmp_seq=4 ttl=64 time=0.215 ms

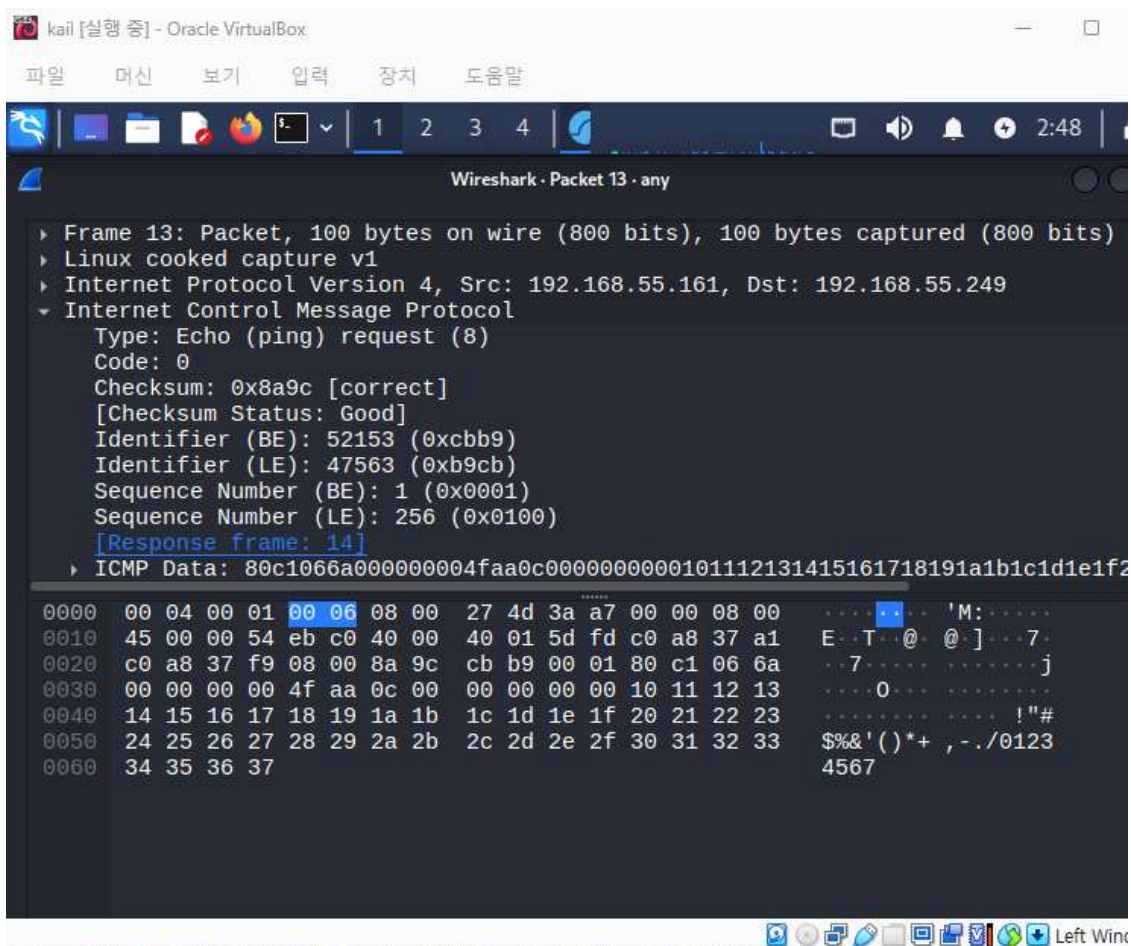
— 192.168.55.249 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3060ms
rtt min/avg/max/mdev = 0.215/0.229/0.256/0.015 ms

(red@kali-Attracker)-[~]
$

hong@NIDS-Suricata: ~
i: device: enp0s3: packets: 72, drops: 0 (0.00%), invalid checksum: 0
hong@NIDS-Suricata:~$ sudo tail -f /var/log/suricata/fast.log
^C
hong@NIDS-Suricata:~$ sudo suricata -c /etc/suricata/suricata.yaml -i enp0s3 -S
/etc/suricata/rules/local.rules
i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
i: threads: Threads created -> W: 1 FM: 1 FR: 1 Engine started.
^Ci: suricata: Signal Received. Stopping engine.
i: device: enp0s3: packets: 44, drops: 0 (0.00%), invalid checksum: 0
hong@NIDS-Suricata:~$ sudo cat /var/log/suricata/fast.log
05/15/2026-06:39:38.550662  [**] [1:2000002:1] [NIDS] SSH Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.55.185:52150 -> 192.168.55.249:22
05/15/2026-06:39:46.521477  [**] [1:2000001:1] [NIDS] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.55.161:8 -> 192.168.55.249:0
05/15/2026-06:39:46.521510  [**] [1:2000001:1] [NIDS] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.55.249:0 -> 192.168.55.161:0
05/15/2026-06:39:55.921521  [**] [1:2000001:1] [NIDS] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0209:74ff:fe8f:db8f:143 -> ff02:0000:0000:0000:0000:0000:0000:0016:0
05/15/2026-06:39:56.128861  [**] [1:2000001:1] [NIDS] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0000:0000:be09:b9ff:fe33:41f0:143 -> ff02:0000:0000:0000:0000:0000:0000:0016:0
hong@NIDS-Suricata:~$
```

사용한 명령어: `sudo snort -c /etc/snort/snort.conf -Q --daq nfq` (Snort NIPS 실행)
`sudo suricata -c /etc/suricata/suricata.yaml -i enp0s3` (Suricata NIDS 실행)
`sudo tail -f /var/log/suricata/fast.log` (탐지 로그 확인)

설명: Snort를 IPS(NIPS) 모드로 동작시켜 Kali Linux로부터 들어오는 ICMP(Ping) 패킷을 실시간으로 차단(Drop)하고 통계를 확인했습니다. 또한 Suricata를 IDS(NIDS)로 구축하여 SSH 스캔 및 핑 공격을 탐지하고 fast.log를 통해 공격 이벤트가 정확히 기록됨을 증명하였습니다.



kail [실행 중] - Oracle VirtualBox

파일 머신 보기 입력 장치 도움말

Capturing from any

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
13	17.144016364	192.168.55.161	192.168.55.249	ICMP	100	Echo (ping) request
14	17.144197737	192.168.55.249	192.168.55.161	ICMP	100	Echo (ping) reply
15	18.144643320	192.168.55.161	192.168.55.249	ICMP	100	Echo (ping) request
16	18.144835462	192.168.55.249	192.168.55.161	ICMP	100	Echo (ping) reply
18	19.168320485	192.168.55.161	192.168.55.249	ICMP	100	Echo (ping) request
19	19.168526608	192.168.55.249	192.168.55.161	ICMP	100	Echo (ping) reply
20	20.192292679	192.168.55.161	192.168.55.249	ICMP	100	Echo (ping) request
21	20.192599752	192.168.55.249	192.168.55.161	ICMP	100	Echo (ping) reply

Frame 13: Packet, 100 bytes on wire (800 bits), 100 bytes captured (800 bits) on interface
Linux cooked capture v1
Internet Protocol Version 4, Src: 192.168.55.161, Destination: 192.168.55.249

0000 00 04 00 01 00 06 08 00 00 00 00 00 00 00 00
0010 45 00 00 54 eb c0 40 00 00 00 00 00 00 00 00 00
0020 c0 a8 37 f9 08 00 8a 9d 00 00 00 00 00 00 00 00

Left Window

```
red@kali-Attracker: ~  
Session Actions Edit View Help  
PING 192.168.55.249 (192.168.55.249) 56(84) bytes of data.  
64 bytes from 192.168.55.249: icmp_seq=1 ttl=64 time=0.224 ms  
64 bytes from 192.168.55.249: icmp_seq=2 ttl=64 time=0.223 ms  
64 bytes from 192.168.55.249: icmp_seq=3 ttl=64 time=0.256 ms  
64 bytes from 192.168.55.249: icmp_seq=4 ttl=64 time=0.215 ms  
  
— 192.168.55.249 ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3060ms  
rtt min/avg/max/mdev = 0.215/0.229/0.256/0.015 ms  
  
(red@kali-Attracker)-[~]  
$ ping -c 4 192.168.55.249  
PING 192.168.55.249 (192.168.55.249) 56(84) bytes of data.  
64 bytes from 192.168.55.249: icmp_seq=1 ttl=64 time=0.191 ms  
64 bytes from 192.168.55.249: icmp_seq=2 ttl=64 time=0.205 ms  
64 bytes from 192.168.55.249: icmp_seq=3 ttl=64 time=0.220 ms  
64 bytes from 192.168.55.249: icmp_seq=4 ttl=64 time=0.320 ms  
  
— 192.168.55.249 ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3048ms  
rtt min/avg/max/mdev = 0.191/0.234/0.320/0.050 ms  
  
(red@kali-Attracker)-[~]  
$ sudo capture-v3  
net: Protocol Version: 4, Src: 192.168.55.161, Dst: 192.168.55.249, Type: Echo (Request)
```

사용한 명령어 `sudo wireshark` (패킷 분석 도구 실행) `ping -c 4 [Target_IP]` (분석용 패킷 생성)입니다

설명: Kali Linux에서 Wireshark를 실행하여 네트워크 인터페이스를 흐르는 트래픽을 캡처했습니다. 생성된 ICMP 패킷의 헤더 정보를 분석하여 Source/Destination IP, Type(Echo Request)등 패킷의 세부 구조와 흐름을 기술적으로 파악하였습니다.

```
mowree@EvilBoxOne: ~  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:3: [hashed name]  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.55.110' (ED25519) to the list of known hosts  
.  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
Connection closed by 192.168.55.110 port 22  
  
(red@kali-Attracker)-[~]  
$ sudo ssh -i id_rsa mowree@192.168.55.110  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
Enter passphrase for key 'id_rsa':  
Linux EvilBoxOne 4.19.0-17-amd64 #1 SMP Debian 4.19.194-3 (2021-07-18) x86_64  
mowree@EvilBoxOne:~$ cat user.txt  
56Rbp0soobpzWSVzKh9YOvzGLgtPZQ  
mowree@EvilBoxOne:~$ whoami  
mowree  
mowree@EvilBoxOne:~$ ls  
user.txt  
mowree@EvilBoxOne:~$
```

john --show key.hash (개인키 비밀번호 크랙) ssh -i id_rsa mowree@[Target_IP] (SSH 침투 접속) whoami & cat user.txt (권한 확인 및 플래그 획득)

설명:타겟 서버(EvilBox)의 취약점을 파고들어 획득한 SSH 개인키를 John the Ripper로 해독하여 패스프레이즈(unicorn)를 알아냈습니다. 이를 이용해 mowree계정으로 원격 접속에 성공하였으며 최종 목표인 사용자 플래스 user.txt값을 획득하며 모의 해킹 과정을 완수했습니다.